

# FingerPet: Enhancing Engagement and Accuracy in Preventive Hand Exercise through Mixed Reality

Zheng Wang

Beijing Jiaotong University

Zichang Ye

Beijing Jiaotong University

Jingya Li\*

Beijing Jiaotong University



Figure 1: (A) Participant performing hand exercises with FingerPet, (B) and (C) show the first-person views, (D) and (E) show the UI interface.

## ABSTRACT

Office workers are prone to hand musculoskeletal discomfort from repetitive tasks. We present FingerPet, an MR application using real-time feedback and a virtual pet to guide preventive exercise. A study with 27 participants showed FingerPet significantly improved outcomes versus conventional methods: intrinsic motivation increased by 66.6% ( $p < 0.001$ ), gesture accuracy by 8.9% ( $p < 0.01$ ), and task completion by 47.5% ( $p < 0.001$ ). Design implications for MR-based exercise and personalization are discussed.

**Index terms:** Mixed Reality, preventive exercise, gamification, hand health, engagement, office worker.

## 1 INTRODUCTION AND MOTIVATION

Office workers are prone to hand discomfort and even work-related musculoskeletal disorders (WMSDs) due to prolonged computer use [1]. Research indicates that preventive exercise can significantly reduce this risk [2]. Yet, prior work on hand exercise has largely focused on rehabilitation and treatment rather than everyday preventive use in workplace [3]. Current autonomous exercise methods often lack real-time feedback to ensure movement accuracy, as well as sustained motivational mechanisms to encourage long-term adherence [4].

To address these gaps, we present FingerPet: a mixed reality application that combines real-time movement feedback with a

gamified narrative layer (see Fig. 1). Designed for autonomous use during short breaks, FingerPet aims to enhance the standardization and engagement of hand exercises, thereby improving exercise quality and frequency to effectively prevent hand discomfort and WMSDs among office workers.

## 2 METHODOLOGY

### 2.1 Hand Exercise Program

FingerPet's exercise protocol is adapted from the validated, image based hand training program by Østerås et al. [5]. We adopted its official illustrated guide as the baseline condition—since no video demonstration is included in the original protocol—to ensure exercise content comparability.

We selected three key gestures (Exercises 4–6) [5] that target hand regions stressed during repetitive office work and are feasible for short, equipment free breaks. Each exercise was set to 5 second holds and 10 repetitions per set—parameters that balance efficacy with time constraints to support daily adherence.

### 2.2 System Design

The gamified narrative centers on a virtual butterfly and garden, designed to map fine grained finger interactions onto a rewarding, emotionally engaging progression. The system is device-agnostic, prototyped on Apple Vision Pro but compatible with any MR hardware. Future lightweight headsets will enable seamless integration of such preventive exercises into daily life.

FingerPet is developed on the Unity platform. It utilizes the native high-definition camera and hand-tracking API of mixed reality (MR) headsets to capture hand joint data in real time, which includes positional information for 21 skeletal points per hand. We have predefined the joint values of standard hand postures in the system and set a certain tolerance range on the basis of ensuring the exercise effect. When the real-time data of the user's hand enters this range, visual and auditory interactions will be triggered, achieving immediate multimodal feedback. The pose-matching logic and feedback engine are implemented in C#

\* e-mail: 10090@bjtu.edu.cn

scripts, with a latency of approximately 12 ms to ensure an immersive and fluid experience. This architecture makes the system compatible with any MR platform that has a hand-tracking SDK, ensuring good replicability.

### 2.3 User Study

We evaluated FingerPet with 27 office workers (22–50 years; 10 male, 17 female) in a within-subjects study. This study followed local ethical standards and participants gave informed consent. Each participant performed the same hand exercises under two conditions: the original picture based protocol and the FingerPet MR system. To preserve ecological validity, participants used their natural sitting posture during the exercises. The order of conditions was counterbalanced, with a 30 minute break between sessions. We collected three types of data:

**Objective metrics:** Since the original protocol has validated the efficacy of the prescribed hand poses, we assessed movement accuracy based solely on pose conformity (not exertion). Hand movements in both conditions were recorded from three fixed high-definition cameras positioned frontally, leftward, and rightward of the participant. The video data were processed offline using MediaPipe [6] Hands to extract the 21 predefined skeletal keypoints per hand. These keypoints were then compared against the corresponding standard pose parameters to quantify accuracy.

**Questionnaires:** The Intrinsic Motivation Inventory (IMI) was administered after each condition. Following prior research [7], we selected four key subscales: Interest/Enjoyment, Perceived Competence, Effort/Importance, and Value/Usefulness.

**Semi structured interviews:** Post session interviews provided qualitative insights into user perceptions and feedback.

### 3 RESULTS

Data were analyzed using paired-sample t-tests in SPSS. Results show that, FingerPet significantly improved exercise outcomes. Specifically, participants' intrinsic motivation increased by an average of 66.6% ( $p < 0.001$ ) (see Fig. 2), gesture accuracy increased by an average of 8.9% ( $p < 0.01$ ), and task completion rate increased by 47.5% ( $p < 0.001$ ).

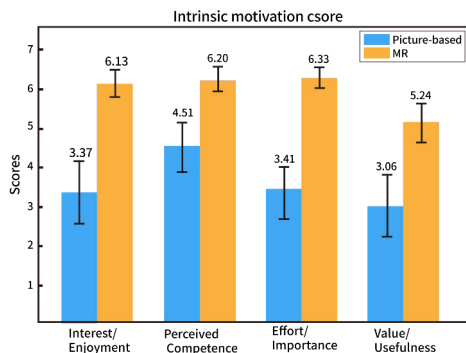


Figure 2: Intrinsic motivation scores of participants under the two hand exercise conditions.

### 4 DISCUSSION AND CONCLUSION

Our study demonstrates that a gamified MR system can simultaneously enhance motivation and movement accuracy in preventive hand exercise. Three core design elements contributed to this outcome: (1) Gamification as sustained motivation transformed repetitive tasks into engaging goal-driven activities; (2) Real-time feedback as quality control provided clear, standardized guidance for self-correction, and (3) Immersion as an

engagement amplifier reduced distractions and fostered focused practice.

This work extends prior MR applications—which often focus solely on either engagement or accuracy—by coupling motivational and feedback mechanisms into a cohesive framework for preventive (non-clinical) use. Meanwhile, a longitudinal study is underway to evaluate the effectiveness of FingerPet in improving long-term adherence. While the study is limited by sample size and short-term evaluation, it provides empirical evidence and design knowledge for creating engaging, autonomous health routines in workplaces.

In conclusion, FingerPet validates an effective approach to reframing preventive exercise from a chore into an engaging MR experience, offering a transferable model for future health and well-being applications.

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### REFERENCES

- [1] F. Gerr, C. P. Monteilh, and M. Marcus. Keyboard use and musculoskeletal outcomes among computer users. *J. Occup. Rehabil.*, 16(3):259–271, Sep. 2006. doi: 10.1007/s10926-006-9037-0.8
- [2] V. Bullo, F. Favro, D. Pavan, A. Bortoletto, S. Gobbo, G. De Palma, S. Mattioli, E. Sala, L. Cugusi, A. D. Blasio, D. C. Diaz, D. S. Bocalini, and M. Bergamin. The Role of Physical Exercise in the Prevention of Musculoskeletal Disorders in Manual Workers: A Systematic Review and Meta-Analysis, 115(1), Feb. 2024. doi: 10.23749/mdl.v115i1.15404.
- [3] T.-W. Kang, J.-H. Lee, D.-H. Park, and H.-S. Cynn. Effects of a Finger Exercise Program on Hand Function in Automobile Workers with Hand Osteoarthritis: A Randomized Controlled Trial. *Hand Surgery and Rehabilitation*, Elsevier, 38(1):59–66, 2019. doi: 10.1016/j.hansur.2018.09.007.
- [4] T. Lima, B. Barbosa, C. Niquini, C. Araújo and R. Lana. Playing against dengue design and development of a serious game to help tackling dengue. 2017 IEEE 5th International Conference on Serious Games and Applications for Health (SeGAH), pp. 1–8, 2017. doi: 10.1109/SeGAH.2017.7939294.
- [5] N. Østerås, K. B. Hagen, M. Grotle, A.-L. Sand-Svartrud, P. Mowinkel, E. Aas, and I. Kjekken. Exercise programme with telephone follow-up for people with hand osteoarthritis—protocol for a randomised controlled trial. *BMC Musculoskeletal Disorders*, Mar. 2014. doi: 10.1186/1471-2474-15-82.
- [6] R. Chauhan, I. Dhyani and H. Vaidya. A Review on Human Pose Estimation Using Mediapipe. 2023 3rd International Conference on Innovative Sustainable Computational Technologies (CISCT), Dehradun, pp. 1-6, 2023. doi: 10.1109/CISCT57197.2023.10351240.
- [7] M. Bressler, J. Merk, T. Gohlke, F. Kayal, A. Daigeler, J. Kolbenschlag, and C. Prahm. A Virtual Reality Serious Game for the Rehabilitation of Hand and Finger Function: Iterative Development and Suitability Study. *JMIR Serious Games*, vol. 12, 2024. doi: 10.2196/54193.